Solution Architecture

Date	5 March 2026
Team ID	
Project Name	SB Stocks – Stock Trading Simulation Platform
Maximum Marks	4 Marks

Architecture Components

1. User (Trader)

The user interacts with the system through the web interface. The user can:

- Register and login to the platform
- View available stocks and market prices
- Buy and sell stocks using virtual money
- Manage portfolios and track performance

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2. Frontend Application (User Interface)

The frontend provides the **interactive interface** for users.

Functions:

- Displays trading dashboard
- Shows stock prices and charts
- Allows buying and selling stocks
- Displays portfolio analytics

Technologies typically used:

- HTML
- CSS
- JavaScript
- React / Web interface

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3. Backend Server

The backend server handles **business logic and application processing**.

Functions:

- Processes user requests
- Handles authentication and authorization
- Executes trading operations
- Manages portfolio updates

Technologies used:

- Node.js
- Express.js

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4. Controller Layer

Controllers act as a bridge between **frontend requests and database operations**.

Responsibilities:

- Validate user inputs
- Process trading actions
- Handle transaction logic
- Send responses back to the user interface

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5. Database (MongoDB)

The database stores all application data.

Stored Data:

- User accounts
- Portfolio details
- Stock transactions
- Historical trading records

MongoDB is used because it provides **flexible schema and fast data retrieval**.

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6. Stock Market Data API

The platform retrieves **stock prices and market information** from external APIs.

Functions:

- Provide real-time stock prices
- Provide historical stock data
- Support stock analysis

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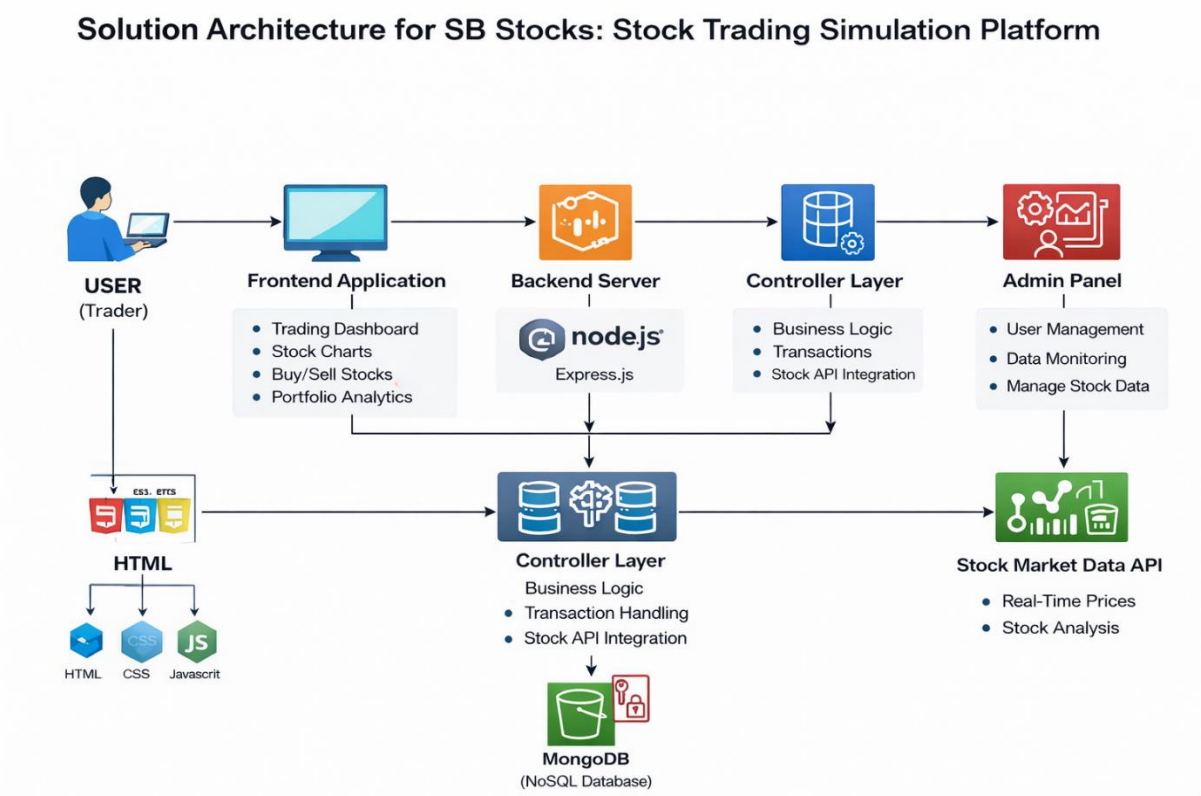
7. Admin Panel

The admin manages the platform operations.

Responsibilities:

- Monitor system performance
- Manage stock data
- Resolve technical issues
- Ensure platform security